INCH-POUND
A-A-52421A(AT)
17 June 2010
SUPERSEDING
A-A-52421(AT)
11 January 1993

COMMERCIAL ITEM DESCRIPTION

ACCESSORY KIT; WARNING DEVICE, DEMOUNTABLE, FOR EMERGENCY VEHICLES

The General Services Administration has authorized the use of this commercial item description (CID) as a replacement for MIL-W-62151(AT) which is canceled.

1. ABSTRACT. This CID covers a 14 volt and 28 volt demountable warning device consisting of a support bar and brackets for rain gutter mounting and gutterless mounting of two rotating warning lights, speaker, electrical harness, electronic siren and public address (P/A) system with microphone for use on security and other emergency vehicles having 12 or 24 volt electrical systems. The accessory kit and its associated hardware is referred to herein as "warning kit".

1.1 Classification. The warning kits shall be of the following types and classes, as specified (see 2.2 and 6.2):

Type I - 14 Volt DC
Type II - 28 Volt DC
Class 1 - Yellow/Amber dome
Class 2 - Blue dome
Class 3 - Red dome

2. SALIENT CHARACTERISTICS

2.1 Materials. Materials shall be of such quality as to be free from defects that might adversely affect performance. Materials having a tendency to warp or to absorb moisture shall not be used. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 3).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online/.

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2.1.1 Painted finish. When specified (see 6.2), a painted finish shall be applied in accordance with detail requirements furnished by the acquisitioning activity.

2.1.2 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

2.2 Design and construction. The warning kit shall come in one of the types and classes specified depending on the voltage and dome color required (see 1.1 and 5.2).

2.2.1 Rotating warning lights. The rotating warning lights shall meet the requirements of A-A-52418. The rotating warning lights shall have a plate completely enclosing the bottom of the housing for mounting onto the bar.

2.2.2 Speaker. A round speaker, having an impedance of between 13 and 16 ohms, shall be supplied with a driver rating of 75 watts (nominal). The speaker shall be weatherproof and corrosion resistant with a mounting plate provided for attachment to the bar.

2.2.3 Mounting bar. The mounting bar with warning lights and speaker shall be in accordance with figure 1. The mounting bar must be mountable on gutterless vehicles or vehicles with conventional rain gutters and it shall be adjustable to accommodate gutters spaced (or roof widths) from 54 inches to 66 inches. All wiring for the warning lights and speaker shall be inside the bar or component with adequate allowances for the full range of bar adjustment. The connector shall extend 1 inch to 4 inches from the end of the mounting bar on the driver's side, and shall be on the rear of the mounting bar. Attaching hardware shall be weather and corrosion resistant, and wire holes in the bar shall be fitted with grommets. Mounting clamps must be tamperproof as far as practicable (employ some method of attachment that discourages vandalism).

2.2.4 Electronic siren and public address system. The siren shall be in accordance with figure 2. Separate on-off switches shall be provided for the siren and warning lights. With the siren switch "ON" either the Alert (yelp), Wail (coaster), HI-LO or P/A (public address system) must be selectable. The microphone shall be of the noise canceling type (magnetic) to limit feedback. The microphone cord shall be retractable and eight feet in length. All control knobs shall be functionally marked. Instructions shall be clearly and permanently displayed on the control unit. All knobs and functional markings on the unit shall be illuminated for night viewing with adequate contrast and brightness. Warning light and siren circuits shall be fused separately. In the public address mode a zero to full attenuation volume control shall be provided. An amplifier connector shall be included to mate with the harness connector in figure 3 (see 2.2.5).

2.2.4.1 Siren tone range. The ratio of the highest fundamental frequency to the lowest shall be 3 to 1. The highest fundamental frequency shall not exceed 1500 Hz.

2.2.4.2 Siren tone modulation.

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a. Wail - The tone range shall be accomplished in a steady rise and fall in 5 to 7 seconds and repeated indefinitely.
b. Yelp - The tone range shall be accomplished in a steady rise and fall in 0.25 to 0.40 seconds and repeated indefinitely.

2.2.4.3 Public address system. Audio frequency response of the public address system shall be flat within ±3 db from 250 to 4000 Hz at rated power.

2.2.5 Harness. The harness shall be as shown in figure 3. A harness connector shall be included to mate with the amplifier connector of figure 2 (see 2.2.4). The power input wire shall conform to SAE J1128 and shall be a minimum of 108 inches long. The ground wire shall conform to SAE J1127 and shall be a minimum of 22 inches long. The cable shall be 0.295 inches outside diameter, jacketed and flexible, with four number 20 AWG shielded conductors. The cable shall be medium duty, 600 volt maximum, oil resistant and heat resistant with low temperature capability.

2.3 Performance.

2.3.1 Rotating warning lights. Rotating warning lights shall meet the performance requirements of A-A-52418.

2.3.2 Electronic siren and public address system. The sound intensity of all siren modes of the system shall be a minimum of 115 decibels at six feet from the speaker on the speaker axis when tested in an anechoic chamber in accordance with ANSI S1.4, ANSI S12.1, ANSI/ASA-S12.51/ISO-3741and ANSI S1.13.

2.3.3 Warning kit. All system functions shall operate with both a 14 volt and a 28 volt power source. All system functions shall operate satisfactorily without interruption during extension of the mounting bar from minimum length to maximum length, and then during contraction from maximum length to minimum length.

2.3.4 Environmental.

2.3.4.1 Vibration. The warning kit excluding the electronic siren and public address system shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 30 Hz. The entire frequency range, from 10 to 30 Hz and return to 10 Hz, shall be traversed in approximately 1 minute. This motion shall be applied for a period of 1 hour in each of three mutually perpendicular directions (total of 3 hours). After the vibration test, the warning kit shall meet the performance of 2.3.3 and shall evidence no permanently deformed, cracked, or broken parts.

2.3.4.2 Extreme temperature. The warning kit shall commence operation within 60 seconds from the time the function of the warning kit is energized after soaking for six hours at both -50±5 °F and 140±5 °F.

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2.3.4.3 Corrosion. The speaker, rotating warning lights, and mounting bar shall be subjected to a 5% salt fog test for 48 hours. After the salt fog test the warning kit shall meet the performance of 2.3.3 and the speaker and mounting bar shall show no evidence of corrosion. The 5% salt solution shall consist of sodium chloride and distilled water, and shall have a pH of 6.5 to 7.2. The exposure temperature for the test item shall be 95±2 °F. The salt fog shall be evenly distributed in the test chamber.

2.3.4.4 Electromagnetic interference. The warning kit shall meet the following emission and susceptibility requirements and limits for commercial equipment mounted on tactical and special purpose vehicles.

2.3.4.4.1 Conducted emission. The warning kit shall meet the requirements of SAE J551.

2.3.4.4.2 Radiated emission. Broadband emissions in the frequency range of 0.15 to 1000 MHz shall not be generated and radiated in excess of the limits shown in figure 4.

2.4 Identification and marking. Unless otherwise specified (see 6.2), identification and marking shall be permanent and legible, and shall include as a minimum the manufacturer's CAGE code number, the warning kit voltage rating, and CID PIN (see 6.4).

2.5 Instructions. The following shall be included with each unit: Complete installation instructions covering assembly, mounting, and interconnection; complete operating instructions; complete maintenance instructions including an electrical schematic diagram; and a parts list for all components.

2.6 Workmanship. Workmanship shall be such as to assure a product free of any malfunction, loose hardware, corrosion, scratches, burrs, sharp edges, chips or foreign materials.

3. REGULATORY REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance

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practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government publications. Copies of A-A-52418 "Light, Warning, Vehicular: Rotating Unit, 14 and 28 Volt DC" are available from the Document Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or https://assist.daps.dla.mil/quicksearch/. The Federal Acquisition Regulation (FAR) may be obtained from www.acquisition.gov/far.

6.1.2 Non-Government publications.

ANSI S1.4	-	Sound Level Meters, Specification for
ANSI S12.1	-	Preparation of Standard Procedures to Determine the Noise Emission from Sources, Guidelines for
ANSI/ASA-S12.51/ISO-3741	-	Acoustics - Determination of Sound Power Levels of Noise Sources Using Sound Pressure - Precision Method for Reverberation Rooms
ANSI S1.13	-	Measurement of Sound Pressure Levels, Methods of

Copies of ANSI documents are available from the American National Standards Institute, 25 West 43rd Street, New York, NY 10036 or www.ansi.org. Copies of ISO documents are available from American National Standards Institute, 11 West 42nd Street, New York, N.Y. 10036 or www.iso.org.

SAE-AS34521	-	Connectors, Receptacle, Electrical, Box Mounting, Rear Release, Crimp Contact, AN Type
SAE-AS34561	-	Connectors, Plug, Electrical, Rear Release, Crimp Contact, AN Type
SAE-AS50151	-	Connectors, Electrical, Circular Threaded, AN Type, General Specification for
SAE-AS85049/1	-	Connector Accessories, Electrical, Backshell, Environmental, Cable Sealing, Straight, Grounding (Without Strain Relief), Category 1C (for SAE-AS50151 Solder Type, V Thread of SAE-AS310X1 Classes A, B, C, or K Connectors)

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SAE-AS85049/139	-	Bushing, Cable Clamp to Cable, Telescoping, for AS85049 Cable Clamps, Category 9
SAE J551	-	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (300 to 1000 MHz)
SAE J1127	-	Battery Cable
SAE J1128	-	Low Tension Primary Cable

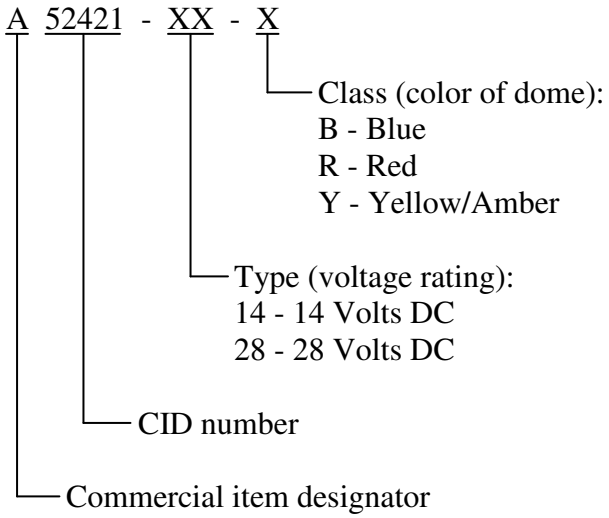
Copies are available from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA, 15096-0001 or website http://www.sae.org/.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 6.1).
c. Type and class of warning kit and PIN (see 1.1 and 6.4).
d. Whether a painted finish is required (see 2.1.1).
e. Marking if other than as specified (see 2.4).
f. Level of preservation, packaging, labeling, and marking (see 4.0).

6.3 Cross-reference data. Warning kits conforming to this CID are interchangeable/substitutable with warning kits conforming to MIL-W-62151(AT).

6.4 Part or identification number (PIN). The PIN to be used for warning kits acquired to this CID is as follows:



6.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified

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tolerances using conversion tables contained in the latest revision of IEEE/ASTM-SI10, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.6 Key words.

Rotating lights
Electronic siren
Public address system

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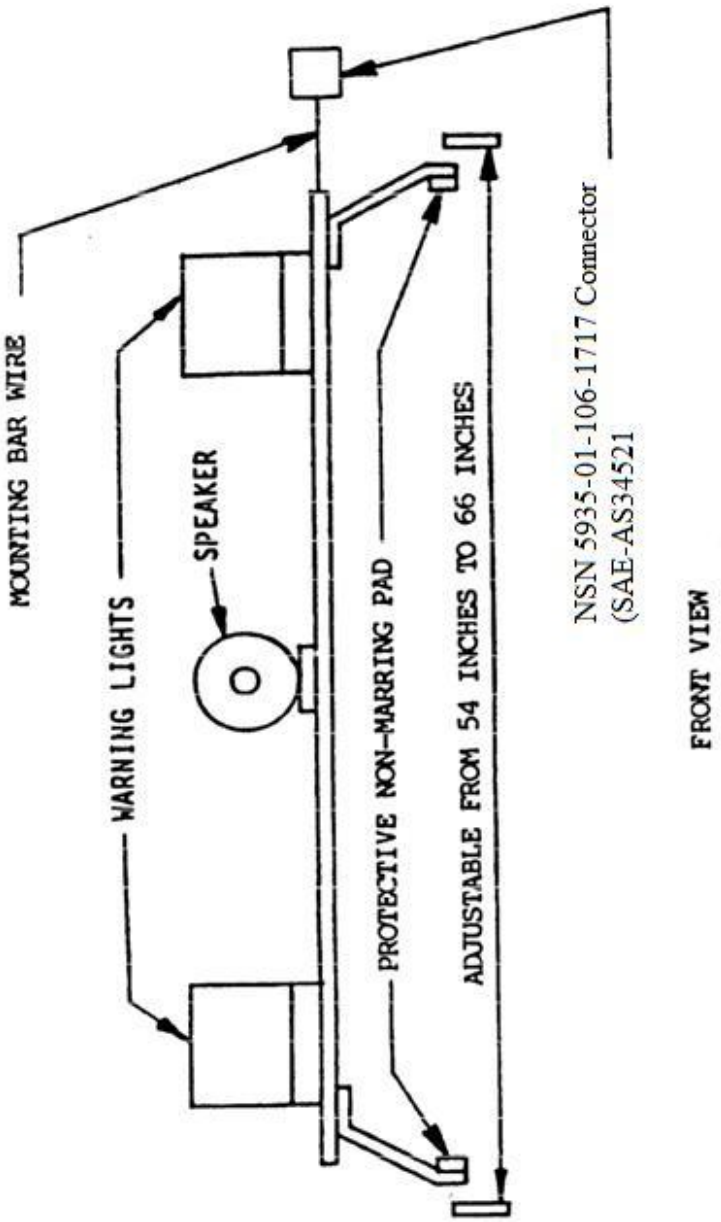
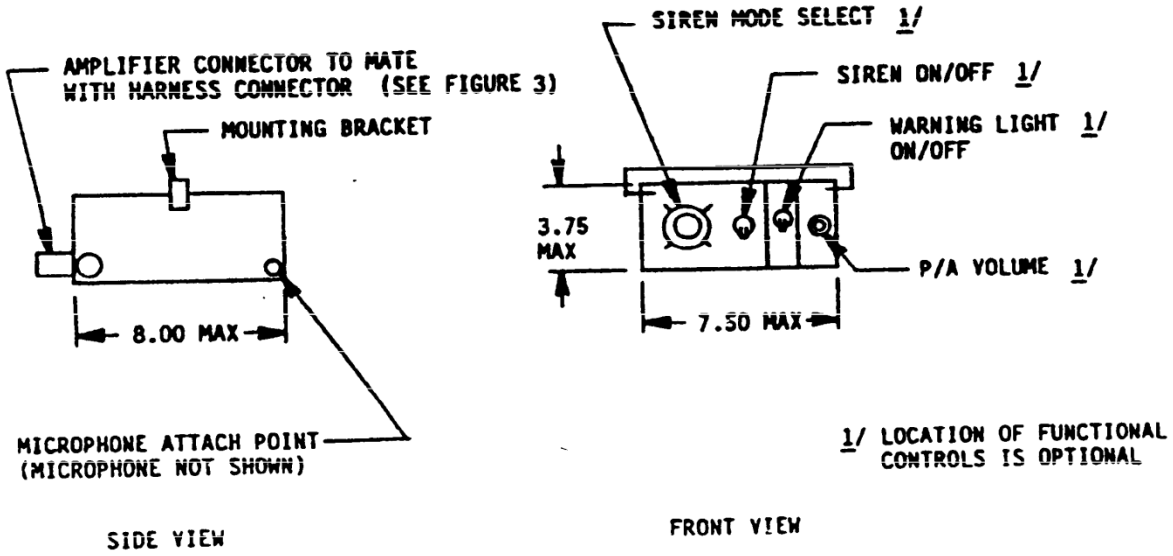


FIGURE 1. Mounting bar with warning lights and speaker.

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NOTE: Dimensions in inches.

FIGURE 2. Electronic siren and public address system.

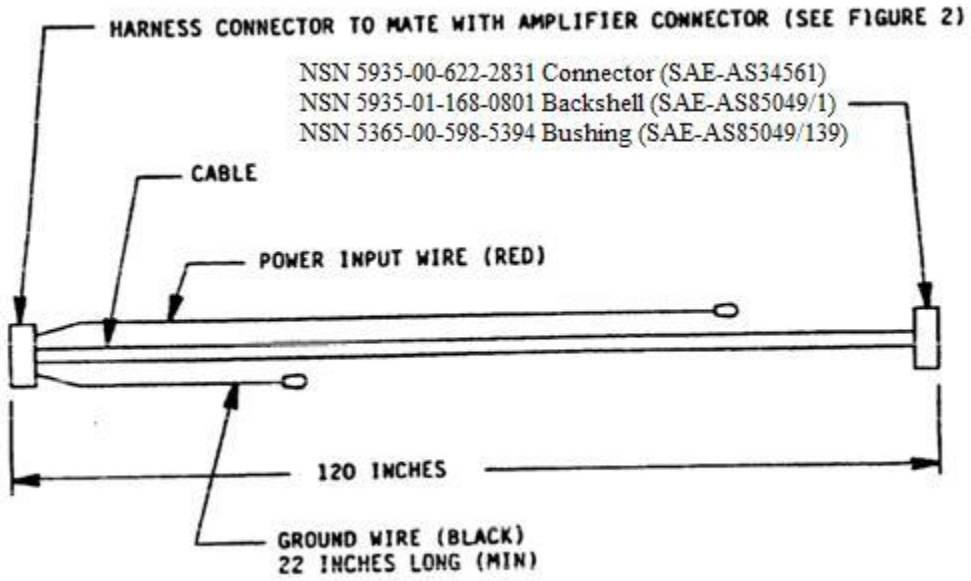


FIGURE 3. Harness.

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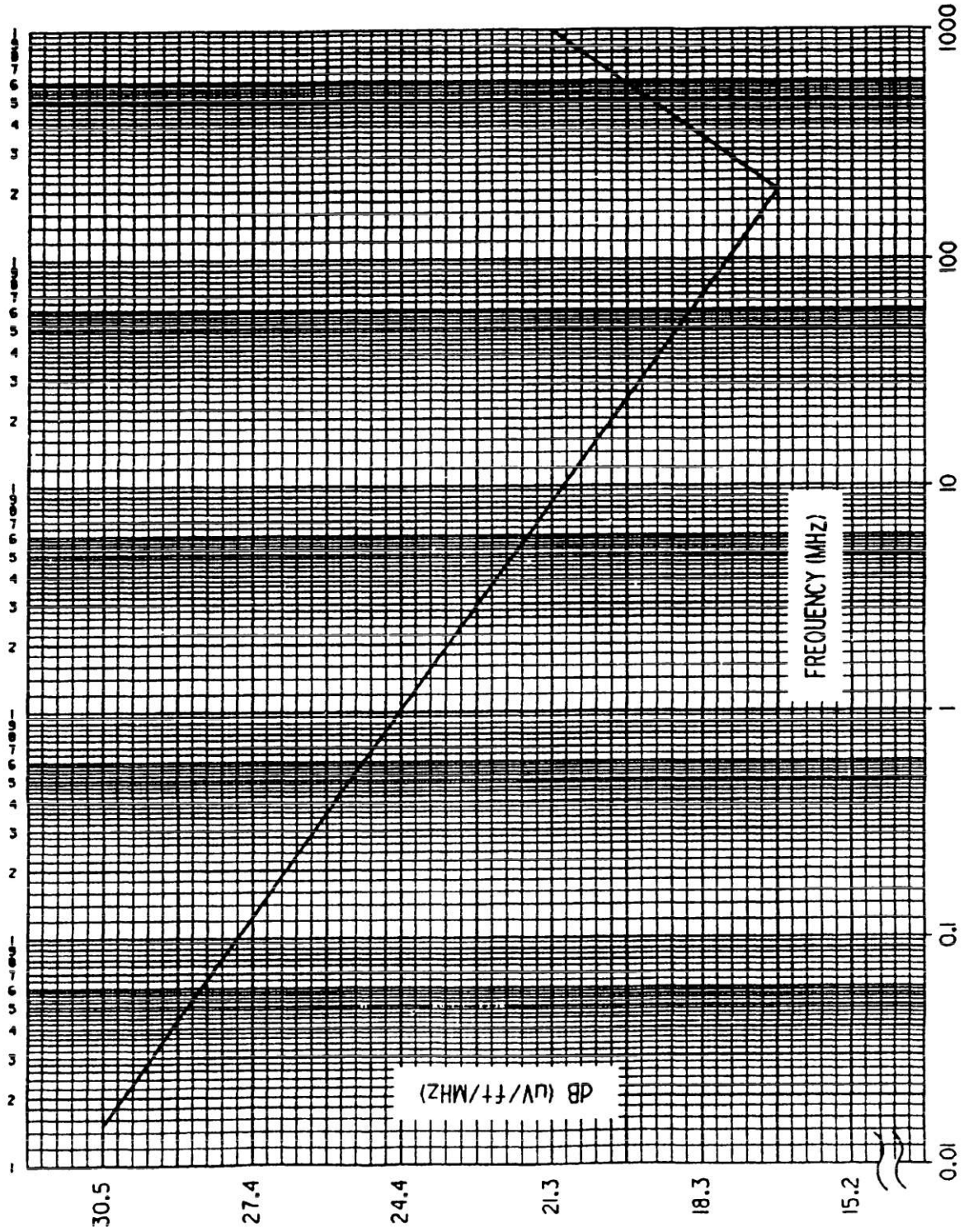


FIGURE 4. Limit for broadband, radiated emissions.

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